Discriminants

Exam code: 9MA0

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Updated on 11 February 2026

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Discriminants

What is a discriminant?

- The discriminant is the part of the quadratic formula that is under the square root sign $b^2 - 4ac$
- It is sometimes denoted by the Greek letter Δ (capital delta)

How does the discriminant relate to graphs and roots?

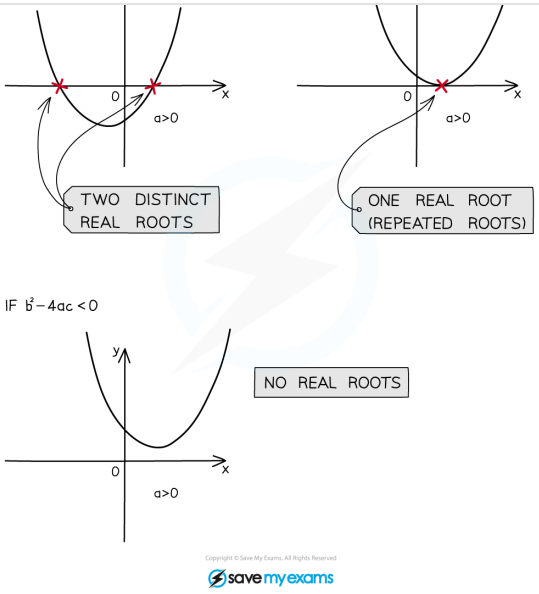
There are three options for the outcome of the discriminant:

- If $b^2 - 4ac > 0$ the quadratic crosses the x-axis **twice** meaning there are **two distinct real roots**
- If $b^2 - 4ac = 0$ the quadratic touches the x-axis **once** meaning there is **one real root** (also called repeated roots)
- If $b^2 - 4ac < 0$ the quadratic **does not cross** the x-axis meaning there are no real roots

Stuck on something?

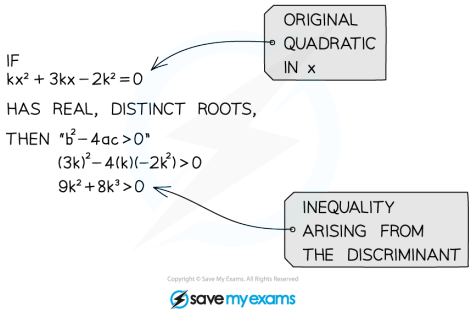
Ask questions and get help with exactly what you're revising.

Got it



Discriminant and inequalities

- You need to be able to **set up** and **solve** equations and inequalities (often quadratic) arising from the discriminant
- Sketch the quadratic and decide whether you're looking above or below zero to write your solutions correctly



Examiner Tips and Tricks

- When questions just mention "real roots", the roots could be **distinct** or **repeated** (i.e. they aren't talking about complex numbers!)
- In these cases, you only need to worry about solving $b^2 - 4ac \geq 0$
- When solving using inequalities, always sketch the quadratic and decide whether you're looking above or below zero to help write your solutions correctly

Worked Example

Find the values of k for which the quadratic function $f(x) = 2kx^2 + kx - k + 2$ has real roots.

$f(x) = 2kx^2 + kx - k + 2$

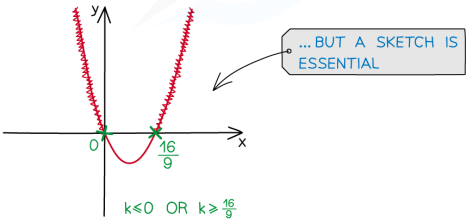
$(k)^2 - 4(2k)(-k+2) \geqslant 0$ ← "$b^2 - 4ac \geqslant 0$" FOR REAL ROOTS

$k^2 - 8k(2-k) \geqslant 0$

$k^2 - 16k + 8k^2 \geqslant 0$

$9k^2 - 16k \geqslant 0$

$k(9k - 16) \geqslant 0$ ← YOUR CALCULATOR MAY SOLVE THIS...



$k \leqslant 0$ OR $k \geqslant \frac{16}{9}$

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